

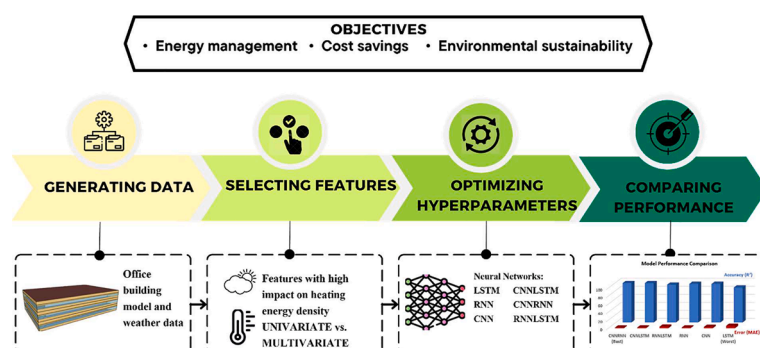
Comparative study of univariate and multivariate strategy for short-term forecasting of heat demand density: Exploring single and hybrid deep learning models

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ABSTRACT

Accurate short-term forecasting of heating energy demand is needed for achieving optimal building energy management, cost savings, environmental sustainability, and responsible energy consumption. Furthermore, short-term heating energy prediction contributes to zero-energy building performance in cold climates. Given the critical importance of short-term forecasting in heating energy management, this study evaluated six prevalent deep-learning algorithms to predict energy load, including single and hybrid models. The overall best-performing predictors were hybrid models using Convolutional Neural Networks, regardless of whether they were multivariate or univariate. Nevertheless, while the multivariate models performed better in the first hour, the univariate models often were more accurate in the final 24 h. Thus, the best-performing predictor of the first timestep was a multivariate hybrid Convolutional Neural Network–Recurrent Neural Network model with a coefficient of determination (R^2) of 0.98 and the lowest mean absolute error. Yet, the best-performing predictor of the final timestep was the univariate hybrid model Convolutional Neural Network–Long Short-Term Memory with an R^2 of 0.80. Also, the prediction accuracy of the best-performing multivariate hybrid models reduced faster per hour compared to the univariate models. These findings suggest that multivariate models may be better suited for early timestep predictions, while univariate models may be better suited for later time steps. Hence,

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combining the models can enhance accuracy at various timesteps for achieving high fidelity in forecasting and offering a comprehensive tool for energy management.

Nomenclature

ANN	Artificial neural network
CNN	Convolutional neural network
CNNLSTM	Convolutional neural network long short-term memory
CNNRNN	Convolutional neural network- Recurrent neural networks
CVRMSE	Coefficient of the variation of the root mean square error
DL	Deep learning
DNR	Direct normal radiation
DHR	Diffuse horizontal radiation
GHR	Global horizontal radiation
GRU	Gated recurrent units

HDD	Heat demand density
KDE	Kernel density estimate
LSTM	Long short-term memory
MAE	Mean absolute error
ML	Machine learning
R^2	Coefficient of determination
RMSE	Root mean square error
RNN	Recurrent neural networks
RNNLSTM	Recurrent neural networks long short-term memory
SVM	Support vector machines
SGD	Stochastic gradient descent

1. Introduction

In cold countries where more than 55 % of total building energy consumption is spent on space heating [1], knowing the building heating energy demand in the next 24 h allows a stable and sustainable energy supply through the optimal use of various energy sources to reduce and shift energy consumption while ensuring that each energy source is used when most effective, and managing the intermittency of renewable ones [2]. For example, thermal energy storage offers a flexible solution for keeping excess energy generated during peak production times and releasing it when the demand is high or renewable energy generation is low [3,4]. The effectiveness of thermal energy storage is enhanced by accurately predicting demand, allowing for more strategic charging and discharging of stored energy [5].

Furthermore, day-ahead forecasting of heat load can enhance the efficiency and operation of district energy networks projected to provide approximately 20 % of global space heating needs by 2030 [6] and, as such, contribute to meeting zero-carbon space conditioning. Moreover, having information about hourly heating energy demand a day ahead may result in substantial economic advantages for smart buildings, as it facilitates the implementation of adapted demand response strategies for reducing peak loads and integrating distributed energy resources [7]. For instance, Lio et al. [8] concluded that optimized energy consumption empowered by accurate short-term forecasting using artificial intelligence-based controls reduced operational costs by around 17 %. Therefore, accurate short-term forecasts of heating demand can contribute to achieving zero-energy building performance and meeting emission reduction targets in cold countries [5].

Nowadays, accurate energy demand forecasting is possible due to a growing recognition of the value of real-time data collection [9]. In this regard, modern buildings are information-intensive due to the implementation of building automation systems [10]. Furthermore, transient hourly and sub-hourly building performance simulation tools for whole-building modeling (e.g., IES, EnergyPlus, eQUEST) have become integral to designing new and retrofitting existing buildings [11]. These accurate whole-building energy models can produce the data sets needed for time series forecasting. In this respect, Gao et al. [11] reported that around 21 % of the published works used datasets generated through simulations to build machine-learning (ML) models. For example, recently Zhang et al. [12] used whole building energy model predictions to train an artificial neural network (ANN) model to forecast heating demand in the United Kingdom with the Coefficient of the Variation of the Root Mean Square Error (CVRMSE) of 14.6 %.

In the literature, different forecasting techniques were utilized,

including ANN models [4,13], applied ensemble methods [14,15], and support vector machines (SVM) [16,17]. These conventional methods face challenges due to nonlinearities caused by occupant behavior patterns [18], uncertainty in building thermal response [19], and variability in weather conditions [20]. In contrast, the remarkable achievements of deep learning (DL) algorithms in resolving intricate statistical challenges and non-linear problems promoted their use for load prediction tasks [21]. In this respect, DL networks may enhance the capabilities of ANN by increasing the number of layers and utilizing their structures to a greater extent, as demonstrated by several recent studies. For example, Suryanarayana et al. [22] showcased the effectiveness of DL models in day-ahead forecasting of heat load for two different district heating networks. An innovative approach to short-term energy load forecasting, leveraging a DL-based method, was also introduced in [18]. The proposed method utilized a wavenet-based model that incorporated a dilated causal residual convolutional neural network (CNN) and Long short-term memory (LSTM) layer to predict hourly load demand in the future. Also, Selvaraj et al. [23] suggested that heat load forecasts based on LSTM networks outperform other methods and have the potential to enhance forecast accuracy further. In [24], the LSTM was used to forecast several Australian households' short-term electricity consumption, and single-horizon forecasts were made for three scenarios: one-hour, three-hour, and six-hour look-back timesteps.

In the context of smart grid implementation, Cai et al. [10] compared traditional time-series models with DL algorithms for building-level load forecasting for a day ahead. They showed that while traditional models have competitive advantages, DL models like gated recurrent neural networks (RNN) and gated CNN can handle high-level uncertainty in building loads. Mariano et al. [24] compared two multi-step ahead prediction strategies with different algorithms to forecast electric consumption in buildings for the next 24 h using shallow and DL algorithms. The DL algorithms performed better when the dataset was extensive, while the shallow learning algorithms performed better when less data was used.

Furthermore, hybrid models have shown excellent performance in forecasting energy. The primary objective of hybrid models is to leverage the benefits of various DL algorithms while mitigating their drawbacks. In this regard, Deb et al. [9] reviewed nine popular machine-learning techniques and concluded that hybrid models were superior for building energy forecasting. Similarly, Sina et al. [25] reported hybrid models' superiority in predicting building energy consumption and expressed the need for more research using hybrid models. Univariate models focus on analyzing and predicting the relationship between a single predictor variable and the target variable,

while multivariate models encompass multiple predictor variables simultaneously. Even though studies show that both modeling methods can provide highly accurate predictions, the comparison between univariate and multivariate models for energy forecasting is less addressed in the literature. One example is Shakeel et al. [26], who developed univariate and multivariate ML models, including LSTM, to test accuracy and reported that the multivariate model showed better accuracy than the univariate model. James et al. [27] compared six univariate models for short-term forecasting of electricity and concluded that the univariate model could handle prediction for 6 h ahead.

The existing literature emphasizes the need for further investigation and performance evaluation of DL models that differ in architecture to provide accurate short-term energy load predictions. Additionally, there is a particular need to showcase the performance of DL models for short-term forecasting of heating energy demand, as most studies ($\approx 47\%$) focused on predicting the total energy demand or ($\approx 31\%$) cooling energy consumption [28].

In this respect, the novelty of this study is a comprehensive evaluation through the comparison of the effectiveness of various DL models, including univariate, multivariate, single, and hybrid, to forecast heating energy demand over a 24 h prediction horizon. This systematic comparison provides a valuable resource for researchers and practitioners seeking the most suitable forecasting approach. Researchers can also build upon these findings by examining more complex hybrid architectures or modifying existing ones. Furthermore, another significant innovation is going beyond the traditional comparisons by exploring optimal hyperparameters for each model, offering guidance on hyperparameter tuning, which is often a critical but challenging aspect of DL model development. This fine-tuning process enhances prediction accuracy and effectively guides practitioners in implementing these models. Moreover, unlike previous studies, this research focuses on accurate predictions and in-depth error analysis, identifying scenarios where models perform well and what type of error they are prone to. This granular investigation provides the knowledge needed to improve the models further, develop strategies for addressing specific challenges in energy forecasting, anticipate potential inaccuracies, and take proactive measures to address prediction errors.

The study aimed to address the identified research gaps and contribute to the field of energy forecasting by exploring the following research questions:

- How do hybrid and single deep DL models differ in their ability to predict multi-step heating loads over 24 h?
- How can DL models be effectively utilized for forecasting heat demand density with a minimal set of features, and what strategies can be employed to maintain high prediction accuracy with limited data inputs?
- Under what specific conditions, including prediction horizon, weather data variations, occupancy patterns, and temporal factors (date and time), do DL models exhibit increased susceptibility to errors in heating load forecasting?

2. Methodology

Fig. 1 illustrates the comprehensive and systematic methodology that has been employed. The methodology is structured into six stages, each playing a crucial role in ensuring the robustness, repeatability, and reliability of the results, outlined as follows:

- Data Generation:** The first stage of the methodology encompassed the generation of heat demand density, the target variable, for the desired commercial building from 2010 to 2019. This process involved running the whole-building energy model of the office building developed by the US Department of Energy (DOE) with EnergyPlus software to produce a time series of the target

variable. Concurrently, data preparation procedures were undertaken, ensuring its quality and consistency.

- Feature Selection:** In the second stage, the wrapper method was used to identify essential features for predicting heating energy demand density. Examples of such input features are dry bulb air temperature and direct normal radiation; selecting a single feature produced univariate models, while selecting multiple features produced multivariate models.
- Data Preprocessing:** The third stage delved into data transformation and segmentation. Data transformation was used to turn a time series problem into a supervised problem using the sliding window technique. Next, data segmentation was used to split the training data from the test data while seeking an optimal balance between the lowest training time and the highest prediction accuracy.
- Model Comparison:** In the fourth stage, potential DL algorithms suited for hourly heat demand density prediction were first analyzed, then run on the training data to produce DL models, and finally run on the test data to compare their predictive capabilities. The combination of multiple DL algorithms together produced hybrid models, which are more sophisticated DL models that can capture more complex patterns, potentially leading to higher prediction accuracy.
- Hyperparameter Optimization:** The fifth stage involved an iterative process for choosing and tuning a set of optimal hyperparameters for the DL algorithms to produce DL models with the highest prediction accuracy. Examples of such hyperparameters are learning rate and number of neurons.
- Performance Evaluation:** The final sixth stage employed a suite of performance evaluation metrics to comprehensively gauge the DL models' predictive capabilities. The predictions generated by the DL models on the training data and test data were compared against the target variable generated in the first stage to quantify their performance. The best DL model for forecasting heating energy load over a 24 h horizon was selected using the ensemble learning technique, "bucket of models" [29].

2.1. Data generation (Stage 1)

This study used the medium office for a building case due to its significant frequency (16 %) in commercial buildings.¹ Obtaining reliable high-resolution (i.e., hourly) heating demand density (HDD) data for multiple years is inherently more challenging than electricity energy use due to the prevalence of manual reading meters and the low penetration of smart gas meters [30]. Therefore, the hourly HDD values were generated using a reference model of the medium office developed by the US Department of Energy (DOE) [31]. Reference models are developed to represent realistic structures and to establish standard or reference energy models for the most occurring commercial building characteristics and construction methods [31]. Furthermore, they are developed in the validated open-source transient modeling tool EnergyPlus [32]. The medium office model has three 4 m high floors with a total floor area of 4851 m² and window-to-wall ratio of 33 %. Each floor has four perimeter zones (40 % of the area) and one core zone (60 % of the area). Fig. 2 depicts the occupancy, lighting, and plug load schedules (i.e., weekdays, weekends, and holidays) used as set points to run the medium office model.

The dataset of 10 recent years (2010–2019) of hourly weather data collected in Winnipeg, Canada [33], was used to run the medium office model and generate hourly heating energy densities, resulting in 86,700 data points of heating demand density. As shown in Fig. 3, Winnipeg experiences extremely low ambient temperatures and, as such, is one of

¹ <https://www.eia.gov/consumption/commercial>

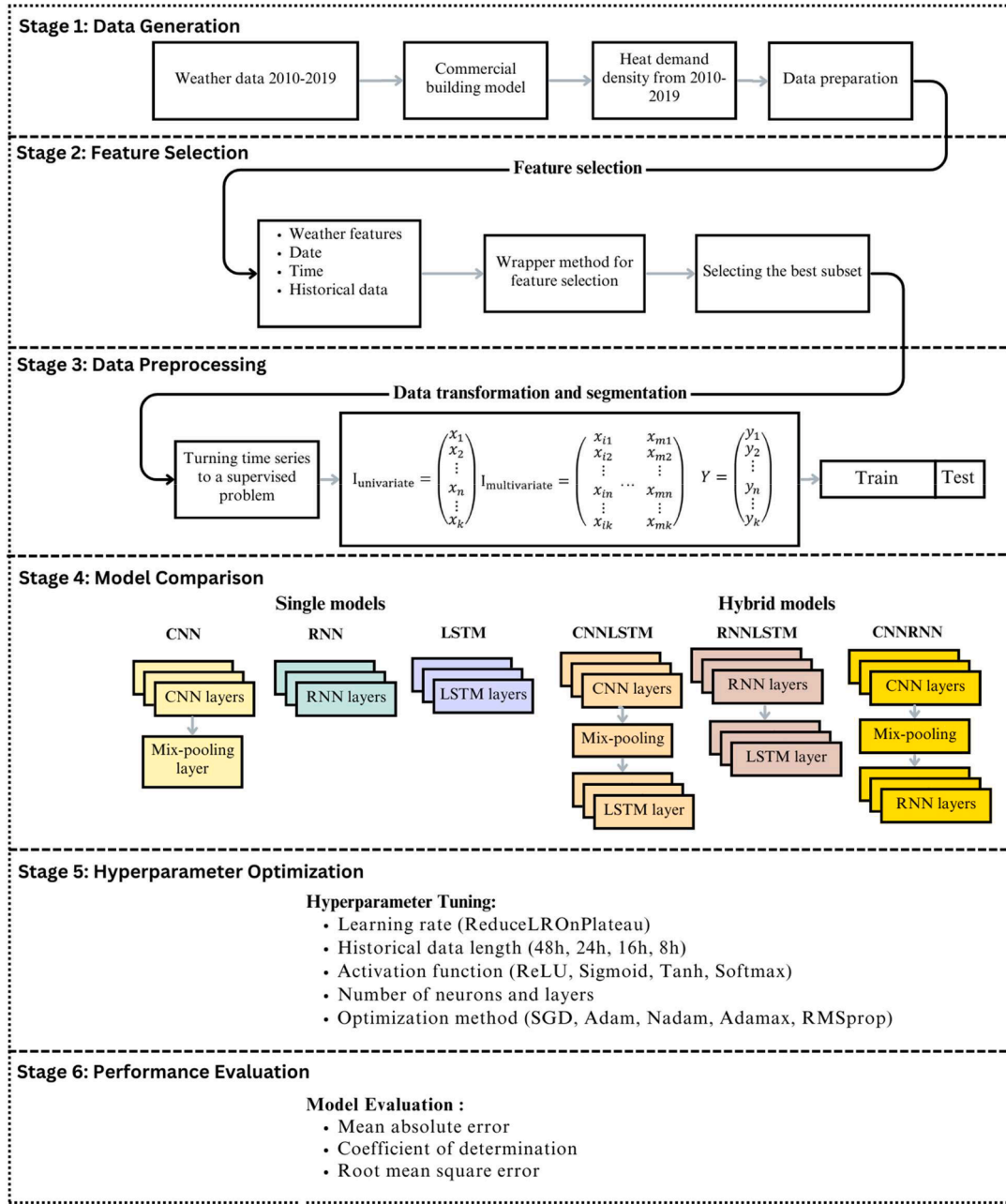


Fig. 1. The six stages of the adopted methodology.

the coldest cities in Canada, with average annual ambient temperature ranging from approximately 3 to 6 °C, resulting in 4500–6000 heating degree days and 10–50 cooling degree days. Hence, annual space heating demand is significantly higher than cooling and needs excellent energy management.

The green line in each box is the median, which is the barrier between Q1 (lower 25 % of data) and Q3 (upper 25 % of data), and each rectangle encapsulates 50 % of data. Based on the median, it can be concluded that the warmest and coldest years are 2010 and 2014.

2.2. Feature selection (Stage 2)

Feature selection is crucial for improving model performance and interpretability. It reduces data dimensionality, mitigates overfitting, enhances generalization, and reduces computational time. Additionally, it identifies critical factors driving predictions, provides insights, and reduces data noise/redundancy [34]. This research used wrapper

methods to find the best features among dry and wet bulb temperature, relative humidity, global horizontal radiation, direct normal radiation, diffuse horizontal radiation, wind direction and speed, snow depth, occupancy schedule, month, day of the week, holidays, hour and previous amount of heat demand density. It can be seen that this method evaluates feature subsets using model performance, iteratively testing and selecting the best combination for accuracy. As such, wrapper methods evaluate DL model performance by adding or removing features sequentially. Hence, this method included a parameter referred to as "k-feature," which represents the desired number of significant features. The examination encompassed three distinct values of k-feature (8, 6, and 4), leading to the evaluation of multiple sets of features. Wrapper methods train and test the model multiple times with different feature subsets, considering feature interactions for accurate selection [35]. While computationally expensive, wrapper methods help improve model performance by selecting the optimal subset of features. Wrapper methods are classified as backward and forward selection. Backward

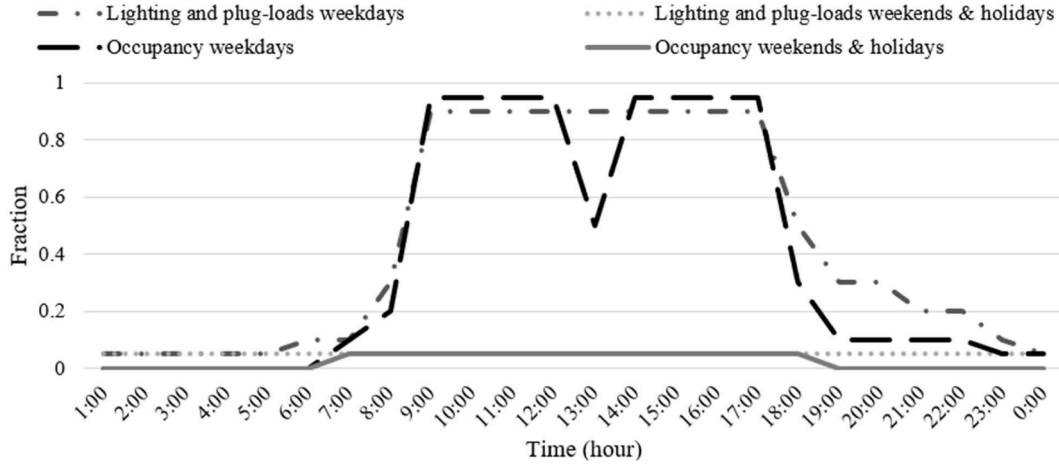


Fig. 2. Occupancy, lighting, and plug load schedules used for data generation (Stage 1).

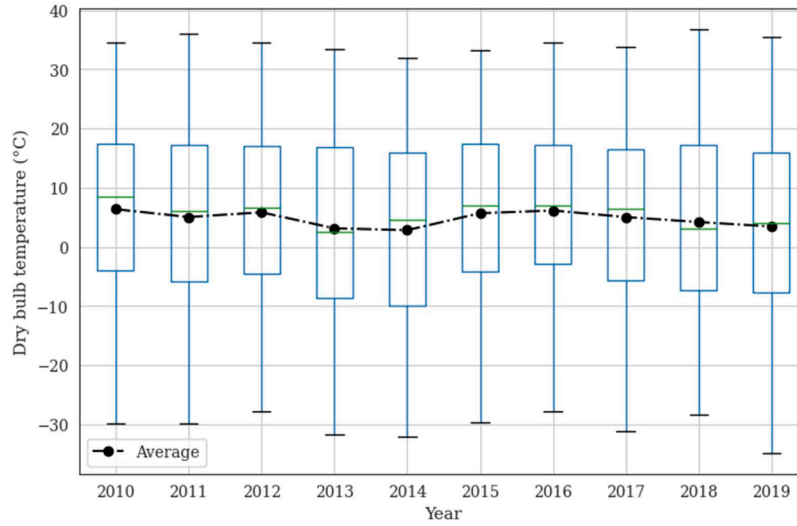


Fig. 3. Annual variability of temperature used for data generation (Stage 1).

selection starts with all features and iteratively removes the one causing the smallest performance decrease until a stopping criterion is met. Forward selection starts with an empty set and adds the feature providing the most significant performance improvement until a stopping criterion is met [35]. This research employed forward and backward feature selection techniques to select the best features. Fig. 4 illustrates the wrapper method used for Feature Selection.

2.3. Data preprocessing (Stage 3)

In this stage, the methodology for converting a time series dataset into a format suitable for multiple-step forecasting is addressed,

complemented by a visual representation in Fig. 5. The X_t shows the dataset at time t . The process starts by converting the time series data into lagged features, where historical data points are used as input variables to forecast future values. This process is achieved through a sliding window technique, where a window of a specified size moves across the time series, creating sequences of input-output pairs. The size of the window is selected to ensure it captures the essential temporal dependencies and patterns inherent in the dataset.

The multi-step prediction involves using multivariate and univariate models trained on historical data, from time step $t-b$ to t . In this context, t denotes the current time, while b denotes the number of prior time steps. Four values for b are analyzed, which are 48, 24, 16, and 8 h.

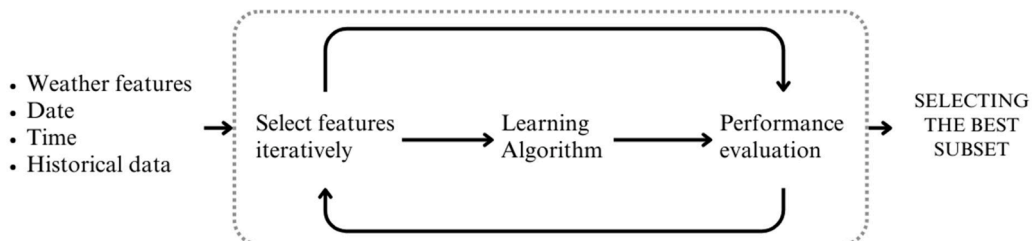


Fig. 4. Wrapper method used for feature selection (Stage 2).

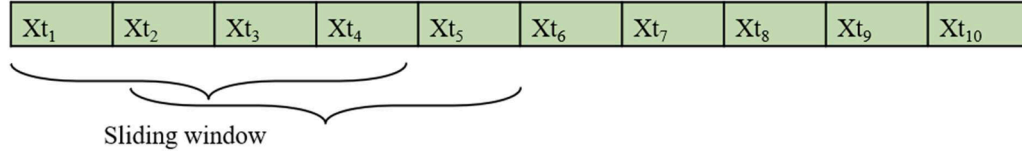
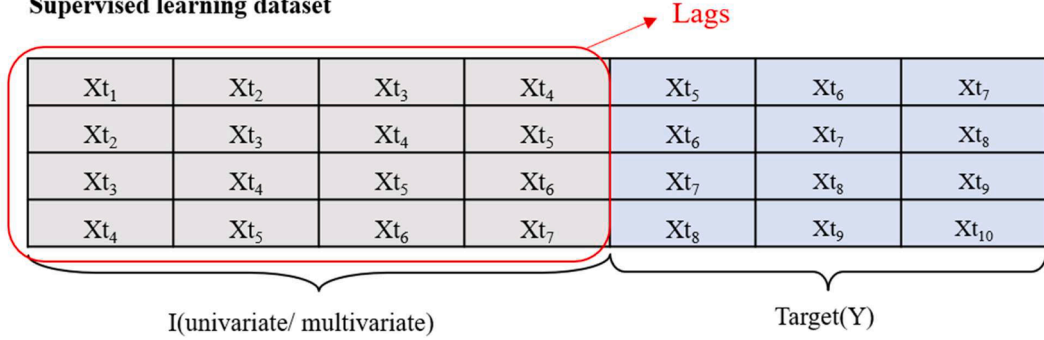
Time series dataset**Supervised learning dataset**

Fig. 5. Sliding window technique used for data preprocessing (Stage 3).

During the initial iteration, the model assimilates the historical data and generates a comprehensive forecast for the subsequent 24 h. The model advances a one-timestep in each subsequent iteration while maintaining a constant data window. Fig. 1 shows that one-dimensional input data ($I_{\text{univariate}}$) was used to feed univariate models, whereas multi-dimensional input data was ($I_{\text{multivariate}}$) where $(x_n - x_1)$ is the length of the training set and $(x_k - x_n)$ is the length of the test set utilized to feed models. Therefore, in the univariate model, only one feature is fed to the model, while in the multivariate m, different features are fed to the model. Nevertheless, the target for both univariate and multivariate is Y. During training, the dataset is windowed based on the length of historical data (b) and the prediction horizon (24 h). The window, which is the sum of the historical data length and prediction horizon, is then moved across the range of data one step at a time, allowing for continuous training and forecasting [36]. The forecasting in the first two windows is as follows:

$$\begin{aligned} F(t+1), F(t+2), \dots, F(t+24) &= \text{model}([I_{t-b}, \dots, I_{t-2}, I_{t-1}, I_t], [Y_{t-b}, \dots, Y_{t-2}, Y_{t-1}, Y_t]) \\ F(t+2), F(t+3), \dots, F(t+25) &= \text{model}([I_{t-b+1}, \dots, I_{t-1}, I_t, I_{t+1}], [Y_{t-b+1}, \dots, Y_{t-1}, Y_t, Y_{t+1}]) \end{aligned} \quad (1)$$

A comparative analysis for data segmentation was applied to determine the most effective approach for forecasting HDD in 2019, considering two main scenarios. In the first scenario, the training dataset spanned from 2010 to 2018, while the testing dataset consisted of 2019. Subsequently, the length of the training data was systematically reduced to evaluate the influence of data length on prediction accuracy. The revised training periods encompassed the years 2011–2018, 2012–2018, 2013–2018, 2014–2018, 2015–2018, and ultimately 2016–2018. This stepwise reduction in training data length allowed for examining how data duration variations impact predictions' accuracy. In the second scenario, three specific years for training were selected, including the warmest year (2010), the coldest year (2014), and a year with mild temperatures (2016), to represent extreme and moderate weather conditions (see Fig. 3), keeping 2019 as the assessment year. These years represent extreme and moderate weather conditions. The goal is to

evaluate how well the model generalizes by training on data from years with distinct weather patterns. Although segmenting data by months, days, or even hours could provide more granular insights into heating demand variations, year-based segmentation provides a reasonable balance between granularity and computational efficiency for time-series data, especially considering heating demand. Also, year-based segmentation captures long-term seasonal trends, which are essential for heating demand prediction, while the heating requirements of buildings vary significantly with changing seasons. Better analyzing these seasonal variations leads to more accurate predictions. By comparing the results of these two scenarios, the study aimed to determine which data segmentation approach yielded superior predictive performance for HDD in 2019. This process allowed for assessing the model's ability to capture seasonal patterns, extreme weather conditions, and the general trend of heat demand in the target year.

2.4. Model comparison (Stage 4)

This study compared the capability of six DL models, including three single CNN, RNN, LSTM, and three hybrid CNNLSTM, RNNLSTM, and CNNRNN, to predict hourly heat demand density (HDD) over 24 h. These models have been selected for their ability to effectively capture temporal dependencies and complex patterns in sequential data, making them valuable tools for various sequential data analysis tasks, including energy-related applications. The Python compiling environment was utilized to conduct all experiments on a machine equipped with an Intel Core i5-11300H CPU. All models are developed through TensorFlow 2.11.1.

2.4.1. CNN

A CNN is commonly used for processing and analyzing data that exhibit a grid-like structure, such as time series data. At its core, a CNN applies filters or kernels to input data in a convolutional layer[37].

These filters are small matrices that slide across the input, performing element-wise multiplications and aggregating to produce feature maps. The filters act as pattern detectors, capturing local patterns or features within the data. By using multiple filters, CNNs can learn and extract complex hierarchical representations of the input data. The convolutional layer is typically followed by activation functions (e.g., ReLU) and pooling layers, which downsample the feature maps to reduce dimensionality and extract dominant features. The extracted features are fed into fully connected layers, where the network learns to make predictions based on the learned representations. The model is trained using backpropagation and optimization algorithms to minimize the prediction error. CNNs can learn meaningful spatial or temporal patterns within the data, such as temperature variations over time or spatial correlations between weather variables and energy. This ability enables CNN to extract essential features for predicting energy based on weather conditions. By training on historical data and corresponding energy values, CNN can learn to generalize and make predictions on unseen data [38].

2.4.2. RNN

An RNN is specifically designed to handle sequential data. In an RNN, each input at a particular timestep is processed along with the hidden state from the previous timestep. The hidden state acts as a memory that captures information from previous inputs in the sequence. This approach allows the network to retain information about past inputs and incorporate it into the current prediction or analysis. The key idea behind an RNN is the application of recurrent connections to enable information flow across different timesteps. These connections allow the RNN to capture dependencies and patterns in the sequential data by learning from the context of previous inputs. However, a limitation of basic RNNs is that they may struggle to capture long-term dependencies due to the vanishing or exploding gradient problem. More advanced RNN variants, such as LSTMs, have been developed to overcome this issue.

2.4.3. LSTM

An LSTM is a subtype of RNN that is specifically designed to capture and analyze sequential data. Unlike CNNs, LSTMs are well-suited for handling the temporal dependencies in sequential data. LSTMs have an inherent memory mechanism that allows them to learn long-term dependencies and capture patterns over extended sequences. This capability is achieved through memory cells, which store and update information at each timestep. At every timestep, an LSTM receives an input and a hidden state from the previous timestep. It then processes the input through several specialized gates, including the forget, input, and output gates. These gates control the flow of information and selectively update the memory cells. The forget gate determines which information from the previous hidden state should be forgotten, while the input gate determines which new information should be stored in the memory cells. The output gate decides which information from the memory cells should be outputted as the current hidden state. By adaptively updating the memory cells and selectively preserving or forgetting information, LSTMs can effectively capture long-term dependencies and remember meaningful contexts over extended sequences [39].

2.4.4. CNNLSTM

The CNNLSTM is a hybrid model combining the strengths of both CNNs and LSTM to process and analyze sequential data effectively. In applied architecture, the CNN layers extract spatial or local patterns from the input data, capturing relevant features across the sequence. The output of the CNN layers is then fed into the LSTM layers, which leverage their ability to model temporal dependencies and capture long-term patterns in the sequence. The CNN layers operate on local receptive fields, using filters or kernels to convolve across the input data, capturing spatial patterns. This approach allows the model to learn

essential features from the input sequence, regardless of their position. The LSTM layers utilize memory cells and gates to remember and selectively update information over time, capturing sequential dependencies and long-term patterns. The CNN-LSTM model can effectively learn and leverage spatial and temporal dependencies in sequential data by combining these two architectures. This feature makes it well-suited for tasks that require capturing both local and long-term patterns [40].

2.4.5. RNNLSTM

The RNNLSTM is a hybrid model that combines the power of RNNs with LSTMs to handle sequential data and capture long-term dependencies. In the proposed architecture, the RNN layers serve as the backbone, processing the input sequence one element at a time while maintaining a hidden state that captures information from previous timesteps. The RNN layers allow the model to retain the memory of past inputs and incorporate them into the current prediction or analysis. The LSTM units within the RNN layers provide an enhanced memory mechanism. They consist of memory cells and gating mechanisms that selectively update and control the flow of information over time. The forget gate determines which past information to discard, while the input gate decides which new information to store in the memory cells. The output gate controls the output of relevant information from the memory cells to the next timestep. By incorporating LSTM units into the RNN architecture, the RNN-LSTM model can effectively capture and model long-term dependencies within sequential data. This approach is beneficial for tasks that require understanding and predicting based on the context of previous elements in the sequence [41].

2.4.6. CNNRNN

The CNN-RNN is a hybrid model that combines the robustness of CNNs and the temporal modeling capabilities of RNNs to process and analyze sequential data efficiently. By leveraging the unique characteristics of both CNNs and RNNs, the CNN-RNN model simultaneously captures spatial patterns and temporal dependencies. In the CNN-RNN architecture, the initial CNN layers extract local or spatial features from the input data. These CNN layers employ filters or kernels to convolve across the input, enabling them to identify and capture relevant spatial patterns. This spatial information is then fed into the subsequent RNN layers. The RNN layers, on the other hand, specialize in modeling temporal dependencies and capturing long-term patterns within the sequence. The RNN layers can retain information over time and selectively update it using memory cells and gates, allowing the model to grasp the sequential nature of the data and uncover intricate temporal patterns. By integrating CNNs and RNNs, the hybrid CNN-RNN model can effectively learn and exploit spatial and temporal dependencies in the sequential data [42].

2.5. Hyperparameter optimization (Stage 5)

The performance of DL algorithms heavily relies on selecting appropriate hyperparameters [43]. Concurrently, DL models often have complex architectures, making it crucial to find the right combination of hyperparameters to achieve optimal and consistent performance. However, performing an exhaustive grid search on all possible hyperparameter combinations can be computationally expensive and time-consuming. Instead, a more efficient approach involves selecting a subset of hyperparameters to tune and updating them based on the gathered information after each run. This iterative process enables making more informed decisions and finding the best hyperparameters for DL algorithms. By employing this method, the hyperparameter optimization process can be streamlined for various DL methods.

The procedure applied involves the following steps. First, the length of training data is analyzed. Then, the historical data length, representing the time window of input data, was examined across four different values: 48, 24, 16, and 8 h. This analysis aimed to determine

the optimal length of historical data that contributes to the model's accuracy and predictive capabilities.

Next, the optimal learning rate is determined using a learning rate scheduling technique called ReduceLROnPlateau [44]. This technique adjusts the learning rate during training based on the validation loss. The analysis encompasses various optimizers, such as Stochastic Gradient Descent (SGD), ADAM, NADAM, ADAMAX, and RMSProp, examining their respective functionalities and efficiencies. SGD is a simple optimization algorithm that updates model parameters by computing gradients on small subsets of the data. ADAM, NADAM, ADAMAX, and RMSProp are advanced optimization algorithms that adapt learning rates based on past gradients, providing faster convergence and better performance for training deep neural networks. The most suitable activation function, ReLU, Sigmoid, Tanh, or Softmax, is chosen. Rectified Linear Unit (ReLU) is an activation function that returns zero for negative input values and the input value for positive values. Sigmoid and Tanh are activation functions that squash input values to a range of (0, 1) and (-1, 1), respectively. Softmax is an activation function used for multi-class classification, producing probabilities for each class that sum to 1. Finally, the ideal number of layers and neurons is determined [45].

2.6. Performance evaluation (Stage 6)

Three metrics, including the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE), were used for the comprehensive evaluation and inter-comparison of the investigated models. The R^2 represents the proportion of the variance in the dependent variable explained by the independent variables. It ranges from 0 to 1, with 1 indicating a perfect fit. The MAE measures the average absolute difference between predicted and actual values in regression tasks. The RMSE provides an estimation of the ability of the model to predict the targeted value; the lower the RMSE, the better the model is as a predictor.

$$R - \text{squared} (R^2) : R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

$$\text{Mean Absolute Error (MAE)} : \text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (3)$$

$$\text{Root Mean Square Error (RMSE)} : \text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}} \quad (4)$$

Furthermore, the ensemble learning technique "bucket of models" [29] was employed to select the best-performing model based on the above performance metrics for predicting the short-term dynamics of heating energy demand. The ensemble learning methods have the following three distinct advantages. First, by leveraging multiple model architectures, the ensemble can capture a wide range of features and temporal dependencies in the data. These results in improved prediction accuracy, especially when compared to single-model approaches that may struggle to capture the full complexity of the data. Second, the ensemble approaches typically enhance accuracy and help identify error-prone scenarios [29]. Comparing the predictions of different models makes it easier to detect situations where the forecasts deviate significantly from the actual values. This insight is valuable for refining and fine-tuning the forecasting process. Third, the ensemble of diverse models offers a holistic approach to capturing temporal data and the excel of each model in recognizing different types of patterns and sequences, thus allowing for a more comprehensive understanding of the underlying dynamics of heating energy demand.

3. Results and discussion

Following the outlined methodology (see Fig. 1), numerous scenarios

Table 1
Results measured during feature selection (Stage 2).

Strategy	Model	Performance metrics	K-feature set
		All steps (1–24)	List of most important features selected by wrapper method
Multivariate 8-features	Single	R^2 0.73	Dry bulb temp, Relative humidity, Direct normal radiation, Occupancy schedule, Wind speed, Snow depth, Day of the week, Heating demand in the previous hour
	Hybrid	0.81	
Multivariate 6-features *	Single	0.75	Dry bulb temp, Direct normal radiation Occupancy schedule, Day of the week Snow depth, Heating demand in the previous hour*
	Hybrid	0.81	
Multivariate 4-features	Single	0.75	Dry bulb temp, Direct normal radiation Occupancy schedule, Heating demand in the previous hour
	Hybrid	0.80	
Univariate 1-feature*	Single	0.69	Dry bulb temp*
	Hybrid	0.77	

* Identifies the optimal k-feature set selected for the following methodology stages.

were examined, yet due to limitations in space, it is not feasible to discuss all of them. Hence, this section focuses on showing the most noteworthy results. These outcomes encompassed diverse aspects, including feature selection techniques, hyperparameter optimization, forecasting results, and sensitivity analysis. The subsequent discussion includes the implications and significance of the outcomes through comprehensive analysis and comparison with the previous studies.

3.1. Results of feature selection (Stage 2)

Table 1 reports averaged R^2 for all step predictions (1–24 steps) of single and hybrid models obtained during feature selection with initial hyperparameters. The findings suggest that increasing the number of features does not lead to better performance of all investigated algorithms, and some features seem to pollute the data and reduce specific algorithms' accuracy. In this respect, the R^2 of single models increased from 0.73 to 0.75 when the number of features reduced from eight to six and four, respectively. In contrast, hybrid models were less sensitive to the number of features than single models, and their accuracy remained at 0.81 for eight and six features and reduced to 0.8 for four features.

Furthermore, the outcomes of backward and forward feature selection techniques were similar, highlighting the significance of dry bulb temperature, direct normal radiation, day of the week, snow depth, occupancy schedule, and heating demand in the previous hour. Considering that these six features achieved the highest average accuracy in all algorithms, they have been selected for training all multivariate models throughout the remaining stages of the methodology. Moreover, it can be seen that dry bulb temperature is the essential parameter appearing in all feature sets, and therefore, it is used to develop univariate models with an accuracy comparable to the best-performing multivariate models. The significance of dry bulb temperature is not surprising, bearing in mind the similar conclusions of other studies [33,46] and the considerable impact of outdoor temperatures on heating energy demand in cold climates [47].

3.2. Results of data preprocessing (Stage 3) and model comparison (Stage 4)

Table 2 compares the prediction accuracy and training computational time with different training datasets. The results are categorized by single models (LSTM, RNN, CNN) and hybrid models (RNNLSTM, CNNLSTM, and CNNRNN) and sub-categorized by univariate models (1

Table 2

Results measured during data preprocessing (Stage 3) and model comparison (Stage 4).

Strategy	Model	Performance metrics								
			2010–2018	2011–2018	2012–2018	All steps (1–24)		2015–2018 *	2016–2018	2010, 2014, 2016
Multivariate	Single	R^2	0.81	0.83	0.83	0.80	0.81	0.83	0.75	0.82
		Time	61 min	57 min	26 min	31 min	26 min	26 min	55 min	27 min
	Hybrid	R^2	0.84	0.84	0.84	0.85	0.83	0.85	0.84	0.85
		Time	87 min	83 min	10 min	12 min	9 min	10 min	15 min	21 min
Univariate	Single	R^2	0.80	0.81	0.82	0.82	0.81	0.83	0.75	0.81
		Time	94 min	80 min	74 min	71 min	65 min	26 min	55 min	43 min
	Hybrid	R^2	0.81	0.81	0.83	0.81	0.82	0.83	0.80	0.83
		Time	50 min	76 min	62 min	46 min	61 min	35 min	41 min	47 min

* Identifies the optimal training dataset selected for the following methodology stages.

feature) and multivariate models (6 features), using the averaged value of R^2 for all step predictions (1–24 steps) and training time for each subcategory. Different training datasets, ranging from nine years (2010–2018) to three years (2016–2018), were considered to assess the models' sensitivity to the training data length. Furthermore, a scenario including the hottest, coldest, and moderate years was also analyzed.

Even though the discrepancy in R^2 values between models is not statistically significant, hybrid multivariate models outperformed the others, with R^2 ranging from 0.83 to 0.85, followed by hybrid univariate models with R^2 between 0.80 and 0.83. The difference between single univariate and multivariate models was even less, with R^2 values for both model types ranging from 0.75 to 0.83. Similarly, Wang et al. [48] forecasted the energy market using univariate and multivariate models and concluded that even though multivariate models outperformed the univariate ones, the difference between them was marginal.

The results also indicate that there is no direct relationship between the length of the training data and accuracy. This observation aligns with outcomes reported in a previous study that forecasted energy use on an urban scale [49]. This study achieved the highest R^2 values, ranging from 0.83 to 0.85, using the training data from 2015 to 2018. Indeed, this training dataset achieves the highest average accuracy for all algorithms and the lowest average training time for all algorithms. Therefore, the 2015–2018 set has been used throughout the remaining stages of the methodology: Hyperparameter Optimization and Performance Evaluation. The second-best scenarios combined the coldest, warmest, and moderate years with R^2 between 0.81 and 0.85. These results highlight the impact of using specific years instead of whole data.

In terms of computational time, it can be observed that the univariate single model required more time than the multivariate. Notably, the longest computational time was observed when utilizing the training data from 2010 to 2018. The univariate model exhibited the highest time consumption, taking approximately 94 min to train. Conversely, the fastest model was found in the multivariate and hybrid approaches, specifically when using the 2014–2018 training data, requiring only 9 min. One of the main reasons for the higher computational time of univariate models compared to multivariate models might be the faster convergence of multivariate models during the learning procedure using ReduceLROnPlateau. The purpose of ReduceLROnPlateau is to reduce the learning rate when the model's performance on a validation metric has plateaued, meaning there is no significant improvement in the performance. This adaptive learning rate adjustment helps to fine-tune the training process, especially when the model approaches a local minimum or faces convergence issues. It can be concluded that the univariate models took more time to converge, and early stop mode was activated in higher epochs, which is intuitive as NNs were forced with fewer features to generalize the data better, requiring more epochs to converge.

3.3. Results of hyperparameter optimization (Stage 5)

As summarized in Table 3, this study also analyzed multiple

Table 3

Values explored during hyperparameter optimization (Stage 5).

Name	Hyperparameter			Performance metrics	
	Value			All strategies and models all steps (1–24)	
	Explored	Initial	Optimal *	Initial R^2	Optimal R^2
Historical data length	8, 16, 24, 48	8	16	0.75	0.81
Activation function	ReLU, Sigmoid, Tanh, Softmax	ReLU	ReLU		
Number of neurons	32, 64, 128	32	64		
Optimization method	SGD, ADAM, Nadam, Adamax, RMSprop	SGD	ADAM		

* Identifies the optimal hyperparameter values selected for the following methodology stages.

scenarios in-depth to identify the optimal setting for historical data length, activation function, number of neurons, number of layers, and optimization method. A comprehensive exploration of these scenarios determined that combining historical data with 16 h, the ReLU activation function and the ADAM optimization method provided the most optimal set of hyperparameters. Also, using 64 neurons and 2 layers in single models and 4 layers in hybrid models provided the most favorable outcomes regarding accuracy and predictive capability. This range of layer numbers and neurons ensured sufficient depth to capture hierarchical representations of the data while avoiding excessive complexity that might hinder generalization. This configuration demonstrated

Table 4

Results measured during hyperparameter optimization (Stage 5).

Strategy	Model	Performance metrics			
		Initial hyperparameters iteration all steps (1–24)		Optimal hyperparameters iteration all steps (1–24)	
		R^2	MAE	R^2	MAE
Multivariate	LSTM	0.69	11.8	0.75	10.7
	CNN	0.77	9.2	0.81	8.6
	RNN	0.78	7.6	0.83	5.9
	RNNLSTM	0.80	6.1	0.82	5.7
	CNNLSTM	0.80	6.2	0.86	5.4
	CNNRNN	0.82	5.8	0.87	4.3
Univariate	LSTM	0.61	12.7	0.65	12.1
	CNN	0.72	10.2	0.79	7.6
	RNN	0.76	9.8	0.82	5.9
	RNNLSTM	0.73	9.8	0.81	6.7
	CNNLSTM	0.78	8.6	0.82	8.3
	CNNRNN	0.81	7.2	0.84	5.8

superior performance, achieving the best accuracy and predictive capability results. Ardabili et al. [25] obtained similar results in which the ADAM optimization method and 2 hidden layers showed the best performance for short-term electricity forecasting. ReduceLROnPlateau is used to adjust the learning rate, and to keep the comparison equitable, the scheduler uses a reduction factor of 0.5 and patience of 5, ensuring that the learning rate never drops below $1e-6$. Based on the results in Table 3, the optimal set of hyperparameters has been used for the final stage of the methodology.

Table 4 shows the effectiveness of hyperparameter tuning in refining the performance of forecasting models using the R^2 and MAE metrics. The univariate RNN model exhibited the most significant predictive accuracy and reliability enhancement. Its R^2 value increased from an initial 0.76 to an optimal 0.82, while its MAE decreased significantly from 9.8 to 5.9. In contrast, the multivariate RNNLSTM model experienced the slightest improvement in accuracy, with its R^2 value modestly rising from 0.80 to 0.82 and its MAE reducing from 6.1 to 5.7. These variations in performance across different models highlight the critical role of customized hyperparameter tuning in optimizing the efficiency of specific algorithmic architectures.

3.4. Results of performance evaluation (Stage 6)

The accuracy of DL models depends on randomness during their training. This study minimized that randomness by setting a consistent seed to focus on enhancing accuracy through hyperparameter tuning. Nevertheless, fixing the seed alone does not guarantee the robustness of the models. Therefore, the minimum, maximum, and standard deviation of R^2 values across 30 iterations presented in Table 5 allowed for analyzing the models' reproducibility and the impact of the randomness on the models' accuracy. It can be observed that multivariate models have higher mean R^2 values ranging from 0.73 to 0.86 and lower standard deviation ranging from 0.0034 to 0.0092 than univariate models with the mean R^2 ranging from 0.64 to 0.83 and standard deviation ranging from 0.0065 to 0.013. Furthermore, the CNNRNN algorithm had the most consistent performance across univariate and multivariate models, showing higher mean R^2 values (0.862) and lower standard deviations (0.0043) than other models. In contrast, the LSTM model displays more significant variability in its predictions, with lower mean R^2 values suggesting less reliability in capturing the underlying variance of the datasets.

Table 6 provides the final comparison of the performance metrics for the six DL models. All results from previous methodology stages have been used for this final comparison, including selected features, optimal

Table 5
Results measured during performance evaluation (Stage 6).

Strategy	Model	Performance metrics			
		30 iterations all steps (1–24)			
		Minimum R^2	Maximum R^2	Mean R^2	Standard deviation
Multivariate	LSTM	0.725	0.749	0.734	0.0092
	CNN	0.798	0.812	0.807	0.0051
	RNN	0.821	0.833	0.828	0.0034
	RNNLSTM	0.804	0.823	0.815	0.0068
	CNNLSTM	0.848	0.859	0.854	0.0038
	CNNRNN	0.856	0.869	0.862	0.0043
Univariate	LSTM	0.623	0.652	0.639	0.0086
	CNN	0.761	0.793	0.780	0.0134
	RNN	0.782	0.819	0.805	0.0119
	RNNLSTM	0.785	0.813	0.80	0.010
	CNNLSTM	0.804	0.823	0.816	0.0065
	CNNRNN	0.819	0.839	0.828	0.0067

* Identifies the best model selected by the "bucket of models" ensemble learning technique.

Table 6

Best results measured during performance evaluation (Stage 6).

Strategy	Model	Performance metrics			Best iteration		
		1st step only			24th step only		
		R^2	MAE	RMSE	R^2	MAE	RMSE
Multivariate	LSTM	0.87	6	12	0.69	10	20
	CNN	0.96	3.3	7.4	0.78	7.7	16.9
	RNN	0.96	1.8	4.2	0.76	7.7	17.2
	RNNLSTM	0.94	3.7	7.7	0.79	7.8	16.6
	CNNLSTM	0.98	1.8	4.2	0.78	7.5	16.8
	CNNRNN*	0.98	1.7	4.3	0.77	7.7	17.2
Univariate	LSTM	0.73	8.5	18.7	0.57	12.3	23.8
	CNN	0.95	4.1	4.4	0.76	7.8	17.3
	RNN	0.96	1.9	4.4	0.78	7.6	16.8
	RNNLSTM	0.93	4.2	4.2	0.76	8.3	17.2
	CNNLSTM	0.96	3	7	0.80	7.5	16.2
	CNNRNN	0.97	2.5	5.8	0.79	7.6	16.5

* Identifies the best model selected by the "bucket of models" ensemble learning technique.

training dataset, and optimal hyperparameters. The comparative analysis reveals that the hybrid CNNRNN models performed the best overall. Hence, the multivariate CNNRNN model predicted the first step with R^2 of 0.98, MAE of 1.7, and RMSE of 4.3. Yet, similar to other models, its accuracy decreased in the 24th hour to R^2 of 0.77, MAE of 7.7, and RMSE of 17.2. The accuracy of univariate CNNRNN was very similar to its multivariate counterpart (see Table 6).

Regarding single models, RNN outperformed the other two models in the first hour, while its performance in the last hour was similar to that of CNN. Additionally, regardless of being trained as univariate or multivariate, the LSTM model demonstrated the poorest performance among the assessed models. In particular, univariate LSTM exhibited the lowest accuracy with R^2 of 0.73, MAE of 8.5, and RMSE of 18.7 in the first step and R^2 of 0.57, MAE of 12.3, and RMSE of 23.8 in the last 24th step.

The results also show better performance of single RNN than hybrid RNNLSTM, especially in the first hour. Such a behavior can be attributed to short-term vs. long-term data dependencies. In this respect, LSTMs are powerful models designed to handle sequences with long-term dependencies. In contrast, RNNs are specialized for short-term dependencies. Therefore, LSTMs might not always be the best choice for every dataset, and a hybrid model with LSTM may even lead to worse performance when the data lacks the specific characteristics that LSTMs excel at capturing. In this regard, when data inherently lacks significant long-term dependencies, using an LSTM might not provide any advantage, and a simple model such as a single RNN may perform better than a more complex hybrid RNNLSTM. It's also possible that the optimal hyperparameters were not appropriate to LSTM. Indeed, the historical data length of 16 h, which was selected for all models, couldn't provide LSTM with data that would exhibit daily or weekly patterns.

3.5. Performance comparison with other studies

As presented in Table 7, the multivariate CNNRNN model developed by this study showcases a compelling performance in short-term energy prediction compared to similar research. For a meaningful comparison, only studies that focused on short-term building energy demand forecasts and used R^2 as a performance evaluation metric are selected for comparison with the CNNRNN model.

Hence, the CNNRNN model, with its robust R^2 value of 0.98 for 1-step and 0.77 for 24-step ahead, outperforms the RNN model for predicting cooling energy in office buildings developed by Chalapathy et al. [50], which had R^2 values of 0.84 for 1-step and 0.61 for the 24-step ahead. The discrepancy in performance might be attributed to differences in the energy type and length of training data. While the CNNRNN model focuses on heating energy in office buildings with a substantial

Table 7

Performance comparison of hybrid multivariate CNNRNN against other studies.

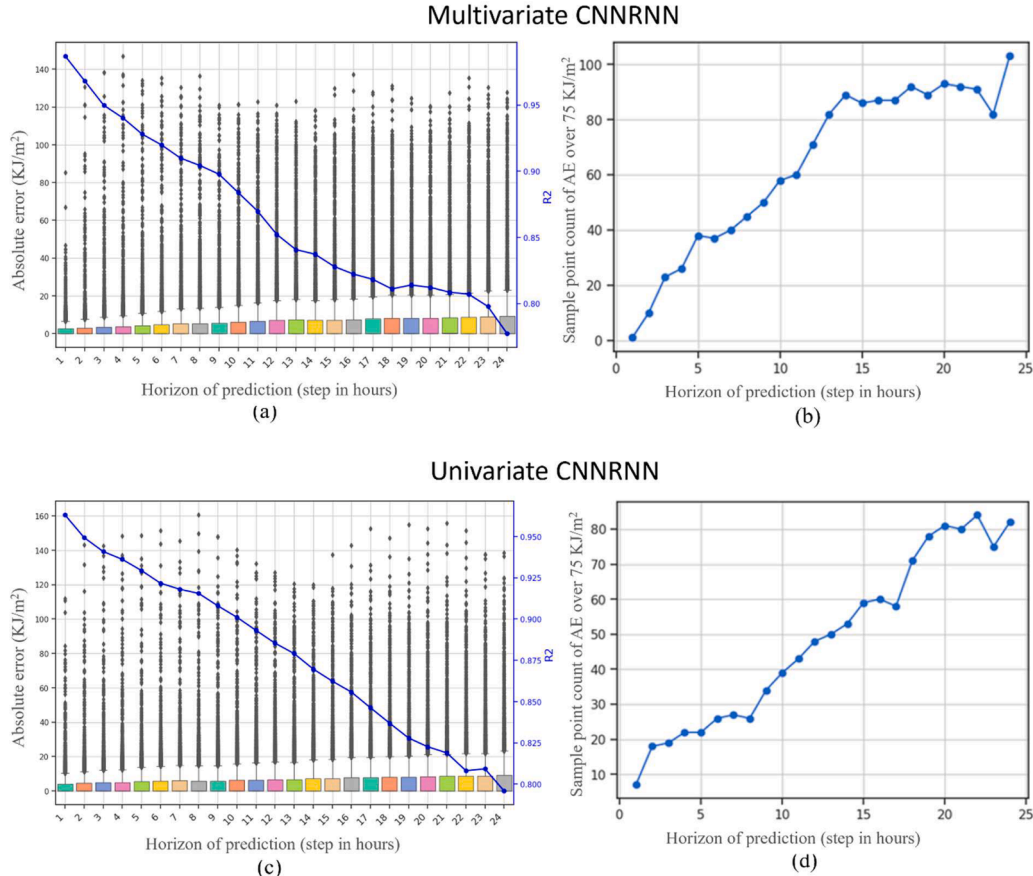
Algorithms	Performance
(This study) CNNRNN	R^2 : 0.77–0.98 Features: Weather data, time, historical load data Energy type: Heating Building type: Office Length of training data: 4 years Prediction horizon: 1–24 h
RNN	R^2 : 0.61–0.84 Features: Weather data, time, historical load data Energy type: Cooling Building type: Office Length of training data: 1 year Prediction horizon: 1–24 h
SVR	R^2 : 0.73–0.88 Features: Weather data, time, historical load data Energy type: Cooling Building type: Office Length of training data: 21 days Prediction horizon: 1–24 h
GRU	R^2 : 0.65–0.95 Features: Weather data, time, historical load data Energy type: Total energy Building type: Six types of buildings Length of training data: 1 year Prediction horizon: 1–12 h
Multilayer perceptron	R^2 : 0.68–0.94 Features: Weather data Energy type: Heating Building type: Commercial Length of training data: 9 days Prediction horizon: 1–24 h

four-year training dataset, Chalapathy's model centers on cooling energy with just one year of training data. The extended training period in this study likely contributes to the enhanced accuracy and predictive reliability. Comparatively, the SVR model by Ding et al. [51] achieves R^2 values of 0.88 for 1-step and 0.73 for 24-step ahead prediction. Like Chalapathy's, this SVR model is trained on cooling energy prediction for office buildings but with a notably shorter training span of just 21 days. This brief period might be a key factor for its relatively lower performance.

The Gated recurrent units (GRU) model by Li et al. [52] presents an interesting comparison. With an R^2 value of 0.95 for 1-step and 0.65 for 24-step ahead prediction, it comes close to the upper limit of the CNNRNN model. The significant drop in R^2 of the GRU model in the last timestep might stem from the model's application across diverse building types, introducing more variability and thus affecting its long-term predictive accuracy. Ding et al.'s model [16], which focuses on heating energy in commercial buildings, achieves an R^2 of 0.94 for 1-step and 0.68 for the 24-step ahead prediction. The steep decline in R^2 value over the 24 h could be attributed mainly to the short training period of 9 days, highlighting the importance of sufficient training data for maintaining accuracy in long-term forecasts.

3.6. Degradation of accuracy by timesteps

Fig. 6 depicts the accuracy of step-ahead predictions from the multivariate and univariate CNNRNN models. The box plots visualize the distribution of absolute errors (AE) for hourly predictions. Selecting a threshold for high absolute error in the context of forecasting heating energy demand density is a subjective decision and depends on the specific application, the acceptable level of accuracy, and the consequences of prediction errors. In this context, the mean heating energy

**Fig. 6.** Degradation of model prediction accuracy by timesteps.

demand density observed at 20 KJ/m² and a standard deviation (STD) of 35 KJ/m², the dataset exhibits a broad range of values from 0 to 200 KJ/m². Given this variability, an AE threshold of 75 KJ/m² is selected. This level focuses on the most significant deviations, particularly during peak demand, which can have substantial operational and financial consequences. This threshold is instrumental in evaluating model performance where peak accuracy is essential, ensuring that the study's insights into model reliability during these critical periods are rigorous and meaningful.

Looking at the outcomes of the multivariate model, as the forecast horizon increases from 1 to 24, the accuracy of the predictions gradually decreases. With longer forecasting horizons, there is an observable increase in variability, indicating higher uncertainty and less precise predictions. In Fig. 6(a) and b), where the multivariate CNNRRN model is used, it can be seen that in 1-step ahead prediction, there is a handful of hours in which AE was beyond 75 KJ/m², while when looking at 20 steps ahead and further steps, the number of sample points can reach around 100. Similar trends for AE and R^2 can be seen in Fig. 6(c) and d), where the univariate model is used. When using the univariate CNNRRN model, the number of hours that AE of 1-step forecasting was beyond 75 KJ/m² was around 10 and reached about 90 when forecasting 24-step. The outcomes suggest that the forecasting model encounters challenges in accurately predicting values as the time horizon extends further into the future. This downward trend in accuracy can be attributed to factors such as error accumulation over time, increased uncertainty in long-term projections, and the influence of unforeseen events or changing dynamics that impact the accuracy of predictions.

3.7. Kernel density estimation by feature

For a closer look at the instances with errors higher than 75 KJ/m², the Kernel density estimation (KDE) for AEs through CNNRRN is calculated. KDE is a non-parametric statistical method for estimating a continuous variable's probability density function (PDF). Hence, Fig. 7

illustrates how often the dry bulb temperature, direct normal radiation, occupancy schedule, day, hour, and month values occurred during the high errors. It can be observed from Fig. 7(a) that the multivariate CNNRRN experienced the highest errors when dry bulb temperature was around -15°C , direct normal radiation was 0 Wh/m², and occupancy schedule was about 0.1. Moreover, it can be seen that high errors occurred when almost no people were inside the building. The univariate CNNRRN model exhibited similar behavior with one significant difference. The dry bulb temperature showed two peaks, at about -20°C and near -10°C . Thus, it can be concluded that the models could not forecast well when the building was almost empty, while their performance improved when it was occupied. Also, Fig. 7(b) shows that both models had high errors when there was no sun or the sky was cloudy (the direct normal radiation was close to zero).

In contrast, their performance significantly improved during days with a clear sky. The likely reason lies in Winnipeg's weather characteristics as it is one of the cities with the highest rate of clear sky in Canada, with almost 30 % of the time experiencing zero Oktas and only 8 % having 8 Oktas which is overcast sky (okta is a unit of measurement used to quantify cloud cover; 0 okta is clear sky with 0 % cloud cover, 1 okta is when 1/8 of sky is covered by clouds) [48]. Thus, the models were trained with insufficient data on cloudy days and may have become more prone to error when predicting cloudy days. Fig. 7(d) illustrates that both models experienced the highest errors on Wednesday, especially multivariate, while Fig. 7(e) shows that the models had the highest errors in the morning around 7–8 am and in the evening (around 7 pm). Finally, Fig. 7(f) shows that both models exhibited the highest errors during winter, particularly in January and February. Considering that January and February are the coldest months with the highest energy peaks, it is no surprise that the models had the highest errors. Another important observation is that univariate and multivariate models present similar error patterns despite generalizing the data differently. As stated earlier, the univariate models took longer to converge with fewer features to generalize the data better. Hence, there

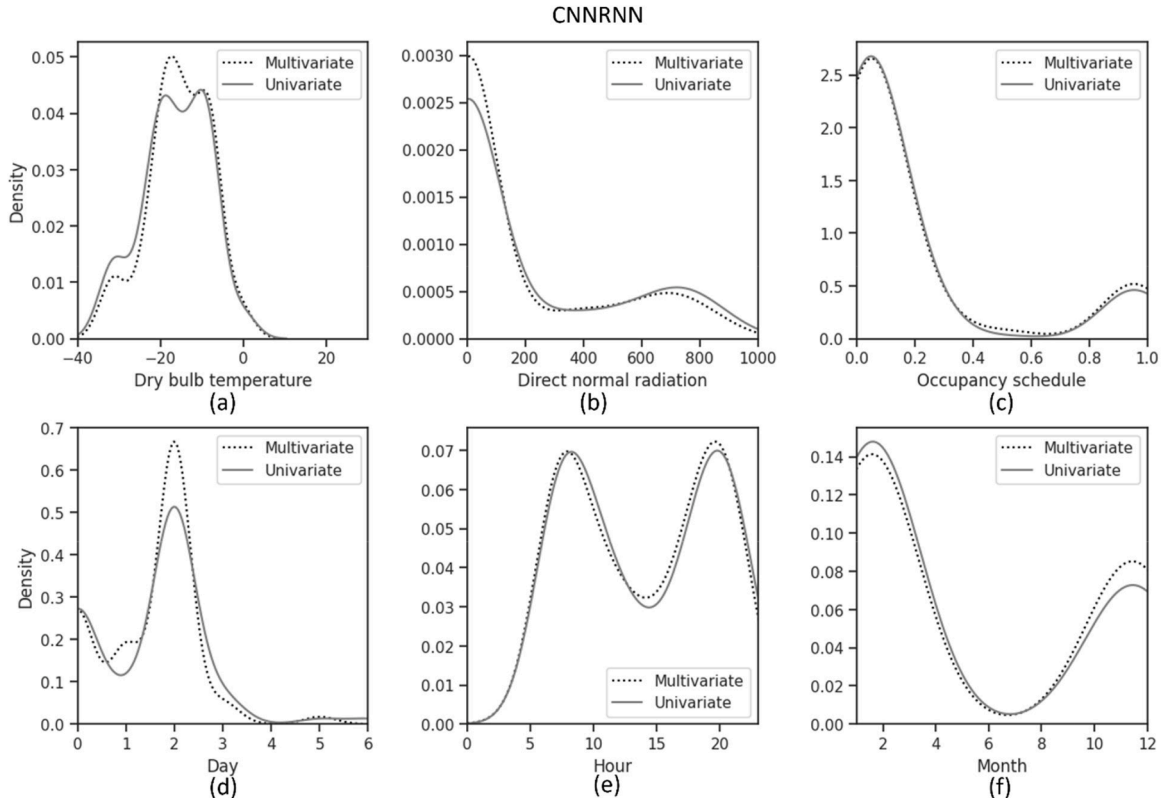


Fig. 7. Kernel density estimation of forecast error distribution over prediction horizon.

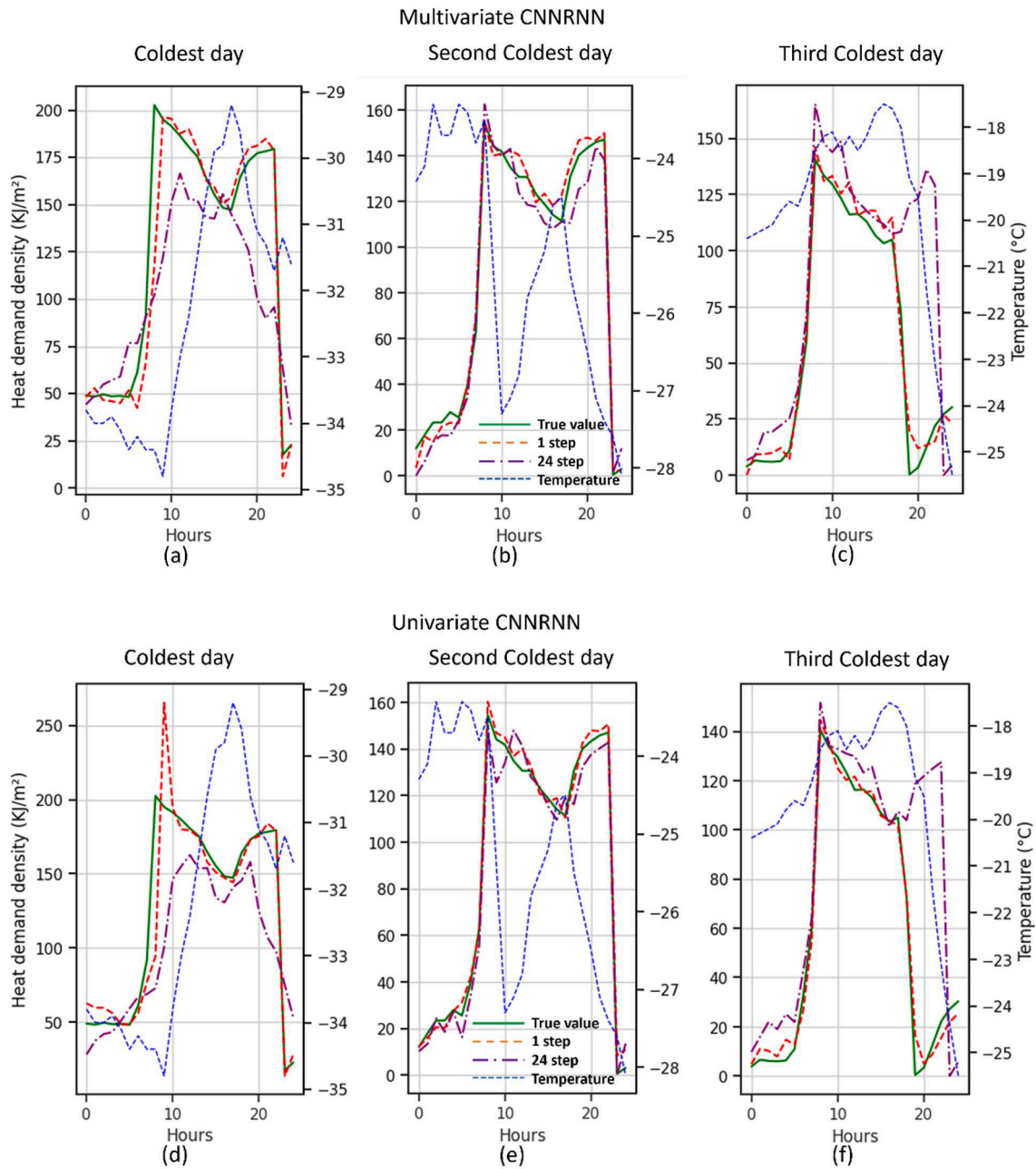


Fig. 8. Multivariate and univariate CNNRNN model predictions on the three coldest days.

is no indication that the multivariate model overfits the data despite converging faster than the univariate model.

3.8. Predicted heating demand density for coldest days

Additional analyses were performed to investigate the ability of the best-performing multivariate and univariate CNNRNN models to predict heat demand density (HDD) during extreme weather events. Hence, Fig. 8 compares the forecasted HDD in the first and last hours of multivariate CNNRNN (a, b, and c) and univariate CNNRNN (d, e, and f) against the simulations of the office building model on the three coldest days. It can be observed that multivariate and univariate CNNRNN models closely follow the predictions in the first hour of the office model during all days. The only exception is the overprediction of the morning peak of the univariate model on the coldest day, as presented in Fig. 8 (d). A possible explanation for this result is the extremely low morning temperature of -35°C combined with the temperature-driven

architecture of the univariate CNNRNN model that does not consider other parameters that could impact HDD, such as solar radiation.

The predictions of HDD in the last hour of both models reduced compared to the first hour, especially on the coldest and third coldest days. In contrast, multivariate and univariate CNNRNN models were able to capture peaks and trends in both horizons on the second coldest day. The likely explanation of these results is the higher representation of a temperature range of -22 to -28°C (second coldest day) than -19 to -25°C (third coldest day) and -29 to -35°C (the coldest day) in the train and test datasets.

3.9. Predicted heating demand density across hours of the day

Fig. 9 shows the annual hourly average of HDD within 24 h forecasted by multivariate and univariate CNNRNN models and their respective forecasted values at a 1-step ahead and 24-step ahead horizon. The findings reveal that the peak in heating demand of

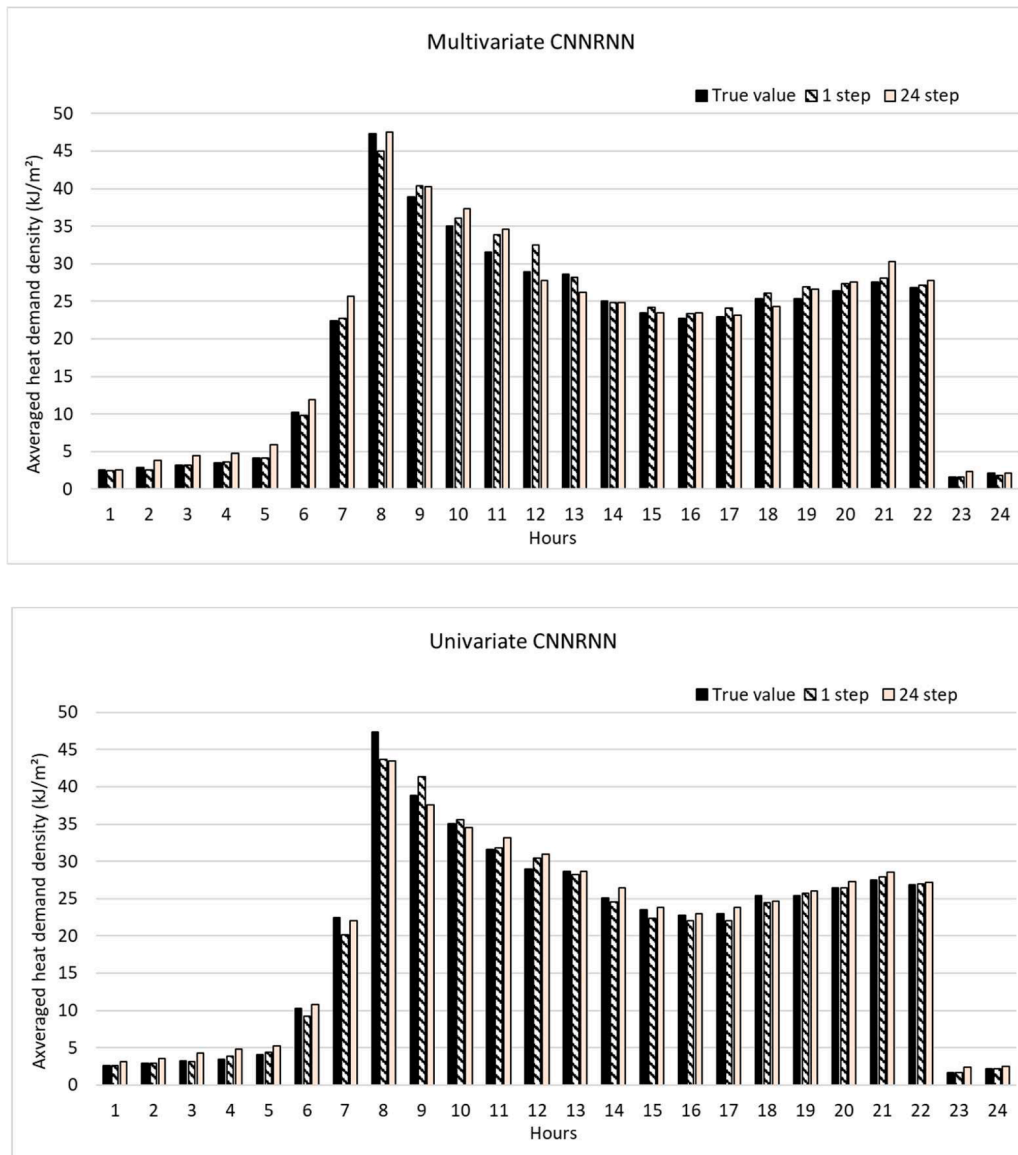


Fig. 9. Comparison of 1-step and 24-step ahead heat demand forecasts across hours of the day.

approximately 48 KJ/m^2 occurs at 7 am, coinciding with the commencement of the heating system to provide a comfortable indoor environment for the occupants. The lowest HDD belongs to 10 pm with values of about 3 KJ/m^2 . Furthermore, the accuracy of the 1-step ahead forecast is notably high every hour, while the 24-step ahead forecast demonstrates lower accuracy and tends to overpredict. Moreover, the multivariate model outperforms the univariate model for predicting HDD every hour, especially at 7 am. A very similar trend can be found in [10], where the authors used several models for forecasting day-ahead energy, and they showed that the highest error occurred at 8 am, and the MAPE was reported as high as 30 %. Also, they show that the lowest errors are from 1 am to 5 am.

4. Conclusion, limitations, and future work

This study is the first to investigate and comprehensively compare the performance differences between deep learning hybrid and single models and multivariate and univariate models in multi-step building heating load prediction over 24 h. Particular focus was placed on examining effective utilization strategies of deep learning models for heat demand density forecasting with a minimal set of input features

while maintaining high prediction accuracy. This encompassing analysis provides valuable insights into the effectiveness of different modeling approaches, aiding researchers and practitioners in making informed decisions. Another novelty of this research is exploring optimal hyper-parameters for each model, a critical and challenging aspect of deep learning model development. Furthermore, unlike other studies, this research evaluated forecast accuracy at several resolution levels (hourly, daily, and monthly) and across various heating demand profiles to thoroughly analyze the robustness of deep learning models.

The findings suggest that the accuracy of hybrid models benefitted to a small extent from the feature number increase, while the opposite happened to single models. Thus, the single model R^2 increased from 0.73 to 0.75 with the decrease in feature number from eight to six and four. On the contrary, hybrid models' accuracy remained at 0.81 for eight and six features and was reduced to 0.8 for four features. The outcomes of backward and forward feature selection techniques indicated that the most significant features were dry bulb temperature, direct normal radiation, day of the week, snow depth, occupancy schedule, and heating demand in the previous hour. Moreover, dry bulb temperature was the essential parameter appearing in all feature sets, and its significance is not surprising considering the considerable impact

of outdoor temperatures on heating energy demand in cold climates.

The results also show that using all training data doesn't guarantee the highest accuracy, and selecting specific years for training can lead to an increase in accuracy compared to the whole data set. Hence, the highest accuracy was achieved when utilizing the training data from 2015 to 2018, with R^2 ranging from 0.83 to 0.85. The second-best scenarios combined the coldest, warmest, and moderate years with R^2 from 0.81 to 0.85.

Furthermore, multivariate models were more effective in forecasting heat demand density than univariate approaches. This outcome highlights the significance of incorporating multiple variables and their interdependencies in predictive modeling. Moreover, this research unveiled the better performance of hybrid models over single models. However, it also concluded that the superiority of hybrid and multivariate models over single and univariate ones was minor, ranging between 2 and 6% in R^2 increase. It was also discovered that using different models for different steps can decrease errors, and combining diverse modeling techniques in hybrid approaches enhances the accuracy and robustness of predictions.

The best-performing hybrid and single models were CNNRNN and Recurrent Neural Network (RNN), with one-step-ahead forecasting accuracy of 0.98 and 0.96, respectively. The CNNRNN and RNN accuracy decreases for subsequent steps, with the 24-step ahead forecast yielding R^2 values of approximately 0.77 and 0.76, respectively. Furthermore, it was discovered that, surprisingly, high errors are more frequent during usual demand, not during the peaks.

In addition, it was observed that all models could forecast the next hour accurately, with R^2 ranging between 0.93 and 0.98, except for the Long Short-Term Memory (LSTM) with R^2 of 0.87. For the 24-step prediction, LSTM exhibited poor performance with R^2 of 0.57, making it an outlier compared to all other models with R^2 ranging from 0.76 to 0.80. It's plausible that LSTM, designed to handle sequences with long-term dependencies, was not performing well because of the short historical data length of 16 h. This length was selected by hyperparameter optimization and applied to all models to ensure appropriate comparison.

The study's limitations that point to the need for future work and investigation are the following. First, in addition to the six algorithms investigated in this study, the short-term predicting capabilities of other forecasting algorithms should be researched. The follow-up study should also include other learning rates and hyperparameter tuning techniques to improve the accuracy and speed of the deep learning algorithms. Second, future work should include more sophisticated ensemble learning techniques such as bagging, boosting, and stacking to reduce the risk of overfitting while improving overall accuracy. For instance, the results demonstrate that specific models are more accurate for specific steps. Hence, a more sophisticated ensemble learning algorithm would exploit the benefits of the ensemble of models differently for each step. Finally, future work should analyze algorithm performance on the heating energy demand data collected from actual buildings typically characterized by more stochastic energy consumption behavior and building types with more irregular energy consumption, such as residential.

CRedit authorship contribution statement

Sajad Salehi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Miroslava Kavagic:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Hossein Bonakdari:** Writing – original draft, Writing – review & editing. **Luc Begnoche:** Formal analysis, Investigation, Methodology, Writing – review & editing.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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